Understanding Large Language Models (LLMs)

Lecture Notes — EduSphere LMS

What is a Large Language Model?

A Large Language Model (LLM) is a type of AI model trained on massive amounts of text data to understand and generate human-like language. LLMs use a neural network architecture called the Transformer, which allows them to process and relate words across long passages of text.

Key Concepts

- **Tokenization** — breaking text into smaller units (tokens) the model can process
- **Embeddings** — representing tokens as numerical vectors capturing meaning
- **Self-Attention** — the mechanism that lets the model weigh relationships between words
- **Pretraining** — learning general language patterns from huge text datasets
- **Fine-tuning** — adapting a pretrained model for specific tasks

How Text Generation Works

1. Input text is tokenized → 2. Tokens are converted to embeddings → 3. Self-attention layers process relationships between tokens → 4. The model predicts the most likely next token → 5. This process repeats to generate full responses.

Real-World Applications

- Conversational AI assistants (e.g. ChatGPT, Claude)
- Code generation and debugging tools
- Content summarization and drafting
- Language translation
- Search and question-answering systems

Next Lesson: We'll explore fine-tuning techniques and how LLMs are adapted for specialized tasks such as coding assistance and domain-specific chatbots.